



Introduction to Photogrammetry

Lecture 12

Geometric Properties of Aerial Photographs

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In Today's Class

1. Introduction
2. Definitions

1. Introduction

- To extract measurements from aerial photographs, we must understand their geometric structure and how objects are projected onto the image.

2. Definitions

Perspective Center

- The perspective center is the location of the camera lens at the time of exposure. All light rays pass through this point.

Focal Length

- Focal length is the distance between the lens and the image plane. It controls the scale and coverage of the photograph.

Principal Point

- The principal point is the point where the camera axis intersects the image plane.

Nadir

- The nadir point lies directly below the camera

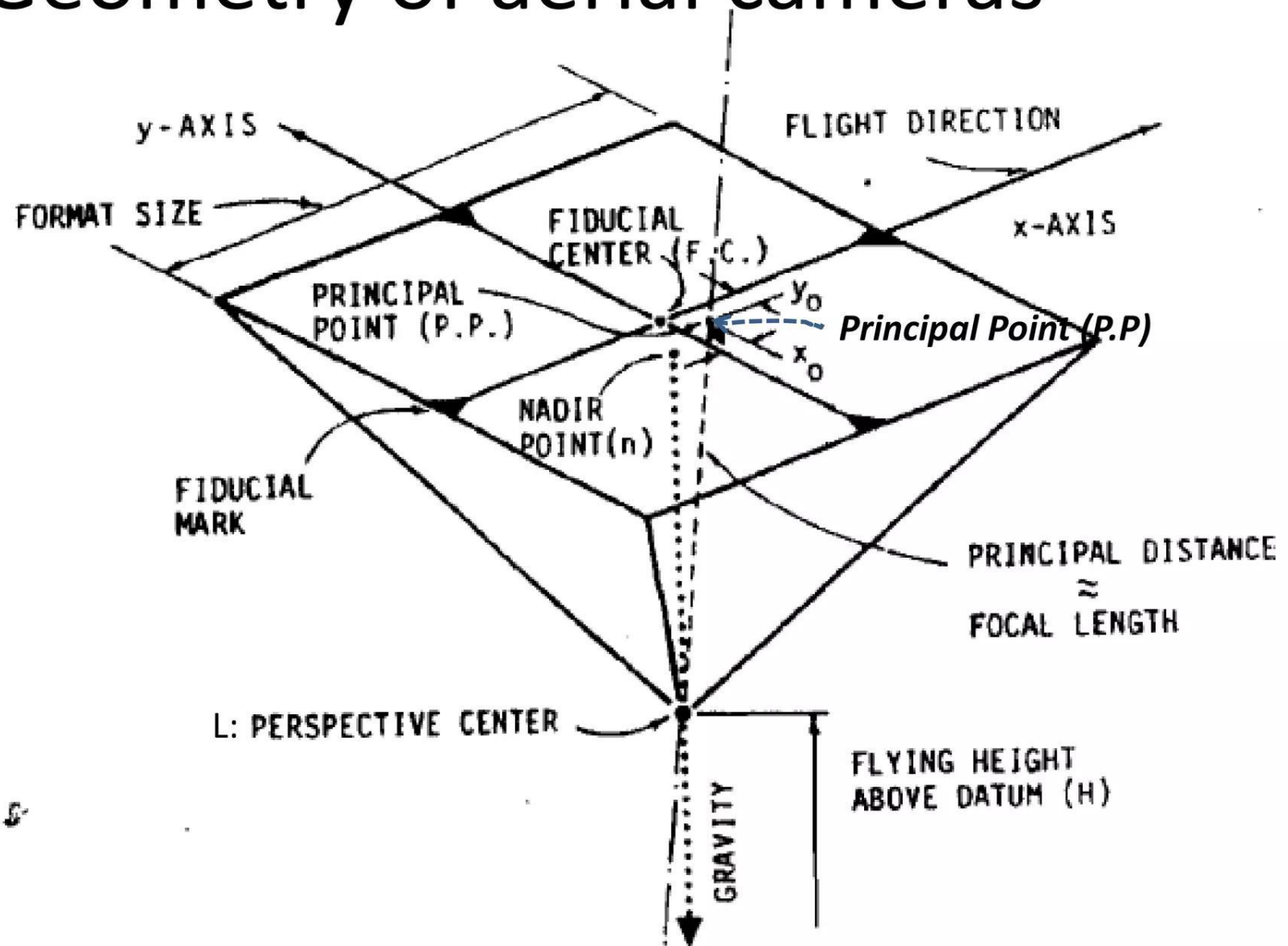
Isocenter

- The isocenter lies between the principal point and nadir

Principal Line

- The principal line connects the principal point, isocenter, and nadir point.

Geometry of aerial cameras



The Principle of Photogrammetry

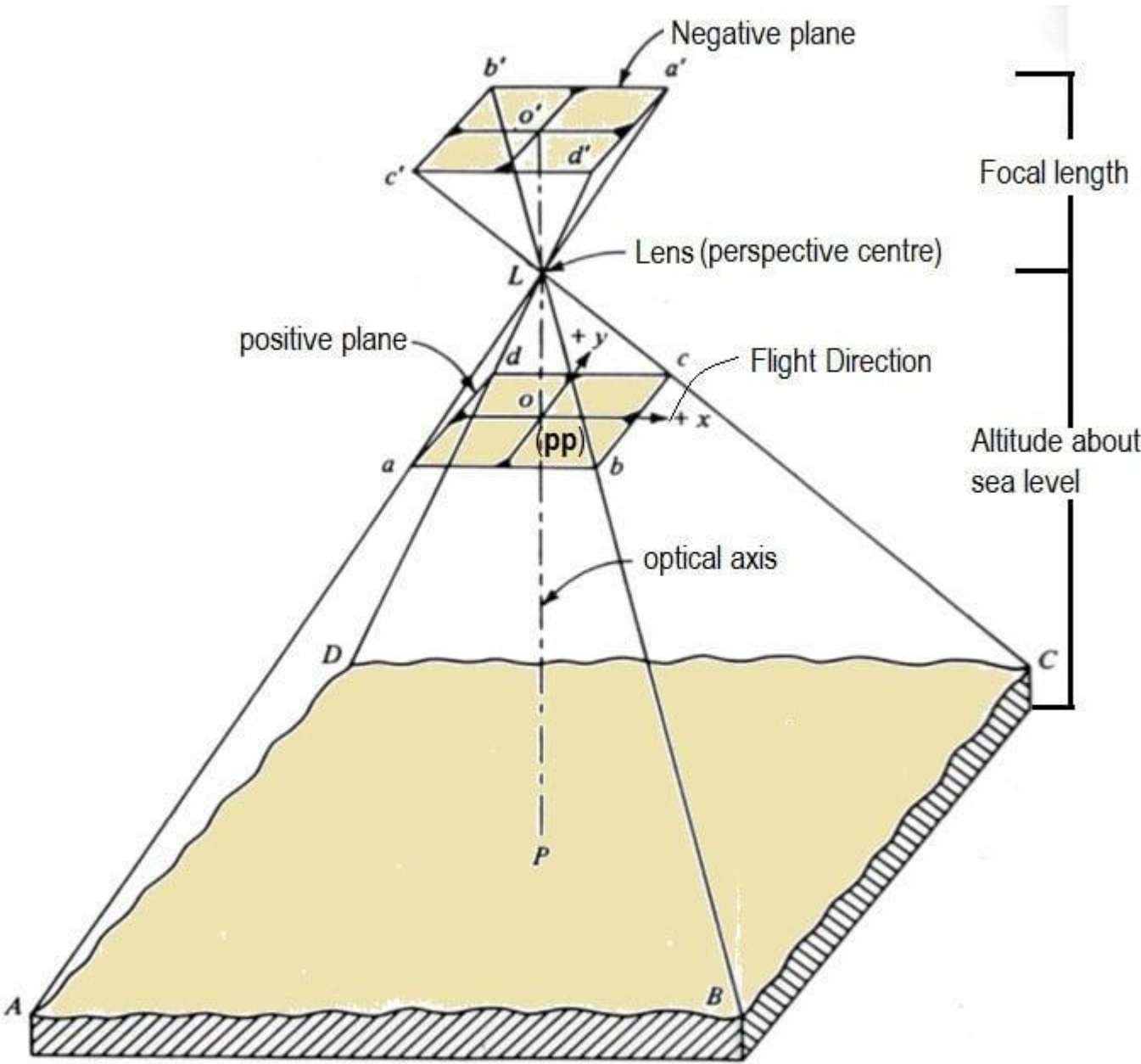
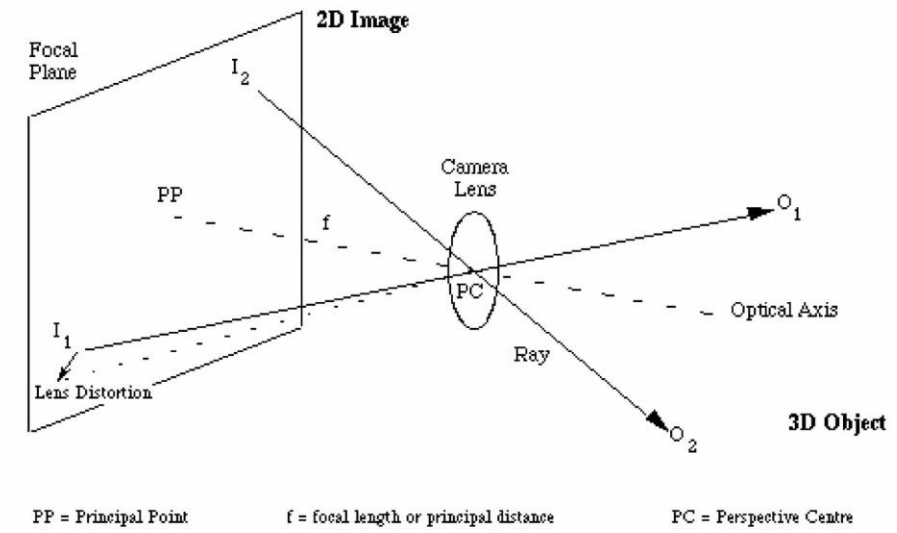


Image and Object Space

- Object space represents the real-world terrain, while image space represents the photograph.

Central Projection

- Aerial photographs are formed using central projection, where all points are projected through a single point onto the image plane.

Photo Scale

- Photo scale is defined as the ratio between a distance on the photograph and the corresponding distance on the ground.
- **Scale Formula**
- $\text{Scale} = f / H$
- Where f is focal length and H is flying height.

- Scale is not constant due to terrain variations and tilt effects.

Scale with Elevation

- Scale = $f / (H - h)$
- This accounts for elevation differences.

Relief Displacement

- Relief displacement refers to the apparent shift of objects due to their height.
- Displacement increases with height
- Displacement increases with radial distance



d

r

h

H

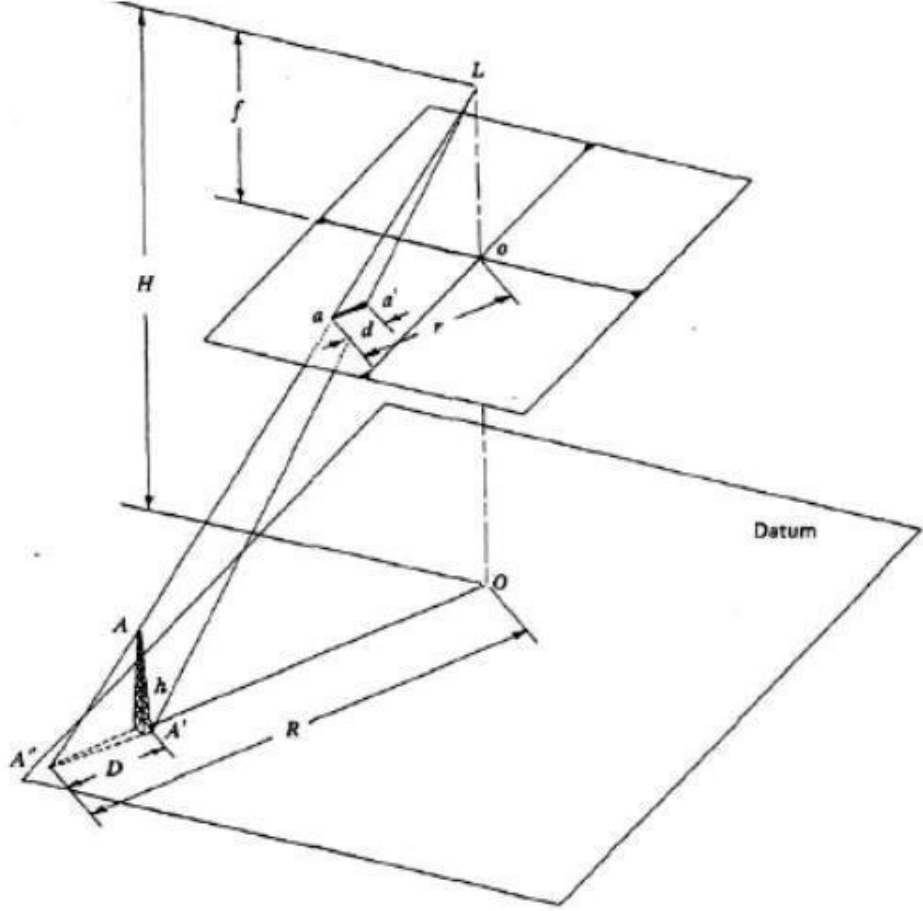


Figure 3.13 Geometric components of relief displacement.



Thank you!

Any Questions
